

ORC · STANDALONE NOTE 2

Quantum Optomechanical Storage of a Single Photon

A Feasibility Analysis and Platform-Specific Noise Budget

*Self-consistent SiN-membrane parameters · Cooperativity $C_0 \approx 2.7$ · $n_{th} \approx 312$ · $\kappa/\omega_m \approx 1.5$
(outside resolved-sideband regime)*

All limitations explained by standard optomechanical theory. No new physics required.

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On the question: can light sit still long enough to see what it does when it isn't flying?

It can — for ~100 ns, with $C_0 \approx 2.7$, in a system that's almost but not quite resolved-sideband. The numbers are the answer.

ABSTRACT

We analyze the feasibility of storing a single optical photon via coherent mapping to a single mechanical phonon in a SiN membrane nano-drum optomechanical cavity. Using a self-consistent parameter set (mechanical mode $Q_m = 10^8$, $\omega_m/2\pi = 1$ MHz, $m = 3.1 \times 10^{-13}$ kg; optical cavity $\lambda = 1550$ nm, length $L = 1$ cm, finesse $F = 10^4$, giving $\kappa/2\pi = c/(2LF) = 1.5$ MHz), the zero-point motion is $x_{zpf} = \sqrt{\hbar/2m\omega_m} \approx 5.2 \times 10^{-15}$ m. The single-photon optomechanical cooperativity is $C_0 = 4g_0^2/(\kappa\Gamma_m) \approx 2.7$. Thermal occupation at 15 mK is $n_{th} \approx 312$, requiring active sideband cooling. However, with $\kappa/\omega_m = 1.5$, the system is outside the resolved-sideband regime ($\kappa < \omega_m$ required), so resolved-sideband cooling formulas do not apply and Doppler-cooling-type limits dominate, giving an effective floor of $n_{min} \gtrsim 1$. Gas damping, thermoelastic damping, and photothermal noise are negligible under cryogenic vacuum and single-photon conditions. The dominant intrinsic loss is support (clamping) loss; phononic soft-clamping projects to $Q_m \approx 10^9$. **We conclude that single-photon storage is not achievable with the baseline parameter set.** The primary obstacles are low single-photon cooperativity ($C_0 \approx 2.7$, below the $C_0 > 10$ threshold for high-fidelity transfer), high thermal occupation, and operation outside the resolved-sideband regime. All limitations are quantitatively explained by standard optomechanical theory; no unidentified loss channel is required. The paper provides a compact end-to-end noise budget for the SiN membrane-in-the-middle platform at telecom wavelength, and a sensitivity analysis identifying ω_m as the highest-leverage parameter for future work.

1. Introduction

The motivating question is direct: can a photonic quantum state be coherently mapped onto a long-lived mechanical degree of freedom? In cavity optomechanics, a photon can be coherently swapped for a mechanical phonon when the single-photon cooperativity $C_0 \gg 1$

and the quantum cooperativity $C_q = C_0/n_{th} > 1$, such that coherent interaction exceeds dissipation and thermal noise. High-fidelity state transfer typically requires $C_0 > 10$ and $C_q > 1$, or the strong-coupling regime $g_0 > \kappa$ and $g_0 > \Gamma_m$. Under these conditions, the photon's energy and momentum can be temporarily stored as a vibration of a nanoscale membrane and retrieved.

We examine a specific implementation: a SiN membrane (nano-drum) inside a high-finesse optical cavity, cooled to 15 mK in a dilution refrigerator. We construct a full loss budget, compute all known noise sources, and show that all limitations are quantitatively explained by standard optomechanical theory. The central conclusion — that single-photon storage is limited by cooperativity and thermal occupation — is well known [1]. The present work provides a quantitative, platform-specific bound for a telecom SiN membrane-in-the-middle system. We emphasize one finding that is structural rather than incremental: the system operates with $\kappa \approx 1.5 \omega_m$, placing it outside the resolved-sideband regime ($\kappa < \omega_m$) required for ground-state cooling. This sets a fundamental ceiling on cooling efficiency that cannot be lifted by parameter tuning alone within the baseline configuration.

2. Self-Consistent Parameter Set

We adopt a baseline drawn from state-of-the-art experiments. All parameters are mutually consistent — in particular, κ is computed from L and F rather than declared independently.

Parameter	Symbol	Value	Justification
Membrane material	—	SiN	Standard, high-Q, CMOS compatible
Thickness	t	50 nm	Typical for MHz mechanical modes
Diameter	D	50 μm	Comparable to optical mode waist
Mass	m	3.1×10^{-13} kg	From density and volume
Mechanical frequency	$\omega_m/2\pi$	1 MHz	Sets thermal occupation scale
Mechanical Q	Q_m	10^8	Below soft-clamped limit $\sim 5 \times 10^8$
Wavelength	λ	1550 nm	Telecom, low fiber loss
Cavity length	L	1 cm	Free-space typical
Finesse	F	10^4	Moderate; avoids excess absorption
Cavity linewidth	$\kappa/2\pi$	1.5 MHz	Computed: $c/(2LF)$, self-consistent
Single-photon coupling	$g_0/2\pi$	100 Hz	Realistic membrane-in-middle value
Bath temperature	T	15 mK	Dilution-fridge base
Cryogenic vacuum	P	10^{-9} mbar	Cryopumping, molecular flow

All derived ratios follow from these inputs. In particular, $\kappa/\omega_m = 1.5$, immediately implying that the system lies outside the resolved-sideband regime (Section 3.7).

3. Noise Budget

3.1 Gas Damping

Residual gas molecules collide with the membrane, extracting energy. Kinetic-theory estimates give $Q_{gas} \gtrsim 10^9$ at $P = 10^{-9}$ mbar. This exceeds Q_m by an order of magnitude and is therefore negligible.

3.2 Support (Clamping) Loss and Two-Level Systems

Energy leaks through the clamped edges into the substrate. Soft-clamped phononic designs [4] demonstrate $Q \sim 10^8$ at room temperature, projecting to $\sim 5 \times 10^8$ at 4 K and $\sim 10^9$ with optimization. The measured baseline $Q_m = 10^8$ sits below this projected limit, indicating additional loss channels — most plausibly two-level system (TLS) defects in the SiN film and at interfaces. Recent measurements [9] suggest TLS losses in stressed SiN at cryogenic temperatures contribute $Q \sim 10^8$ – 10^9 depending on strain configuration, consistent with the gap between baseline and soft-clamped limits.

3.3 Thermoelastic Damping

Thermal gradients across the membrane produce strain dissipation. At cryogenic temperatures, the heat capacity scales as $C_v \propto T^3$ (Debye), strongly suppressing TED. For SiN at 15 mK, $Q_{\text{TED}} > 10^{10}$. Negligible.

3.4 Photothermal Heating

A single 1550 nm photon carries $E_{\text{ph}} = hc/\lambda \approx 1.28 \times 10^{-19}$ J. The cavity storage time is $\tau_{\text{cav}} \approx 1/\kappa \approx 1.06 \times 10^{-7}$ s, giving an average power of $\sim 1.2 \times 10^{-12}$ W. Even at 10 ppm absorption, the absorbed power is $\sim 10^{-17}$ W — far below any thermally relevant scale. Photothermal effects are negligible.

3.5 Zero-Point Motion and Single-Photon Displacement

The relevant displacement scale is

$$x_{\text{zpf}} = \sqrt{\hbar/(2m\omega_m)} \approx 5.2 \times 10^{-15} \text{ m}.$$

The static displacement induced by a single intracavity photon is $\Delta x \approx 2g_0 x_{\text{zpf}}/\omega_m \approx 1.0 \times 10^{-18}$ m. This is well below shot-noise-limited interferometric sensitivity; readout therefore relies on the cavity frequency shift $\Delta\omega_{\text{cav}} = g_0$ rather than direct displacement detection.

3.6 Thermal Phonon Occupation

The equilibrium phonon number is $n_{\text{th}} = (e^{\hbar\omega/kT} - 1)^{-1}$. For $T = 15$ mK and $\omega_m/2\pi = 1$ MHz, $\hbar\omega_m/k_B T \approx 0.0032$, so the high-temperature limit applies:

$$n_{\text{th}} \approx k_B T / (\hbar\omega_m) \approx 312.$$

Raising ω_m to 10 MHz at the same temperature would reduce n_{th} to ~ 31 . Achieving $n_{\text{th}} < 1$ (passive ground state) at 1 MHz would require $T < 50$ μ K, far below standard dilution-fridge capability. Active cooling is therefore mandatory.

3.7 Cooperativity and the Sideband Regime

The single-photon cooperativity is

$$C_0 = 4g_0^2/(\kappa\Gamma_m), \Gamma_m = \omega_m/Q_m.$$

With $g_0 = 628$ rad/s, $\kappa = 9.42 \times 10^6$ rad/s, $\Gamma_m = 0.063$ rad/s:

$$C_0 \approx 2.67, C_q = C_0/n_{\text{th}} \approx 0.0086.$$

Both numbers fall below the standard high-fidelity thresholds: $C_0 > 10$ for clean single-photon swap, $C_q > 1$ for coherent interaction to dominate thermal noise.

Sideband regime. The system operates with $\kappa/\omega_m = 1.5$, just outside the resolved-sideband regime ($\kappa < \omega_m$ required for clean ground-state cooling). The ideal-limit formula $n_{\text{min}} = (\kappa/4\omega_m)^2 \approx 0.14$ does *not* apply here because it assumes resolvable sidebands. Outside that

regime, sideband-cooling efficiency is fundamentally compromised: red and blue sidebands overlap, limiting cooling to Doppler-cooling-type performance with $n_{\text{min}} \geq 1$. Achieving true ground-state preparation requires either reducing κ (higher finesse) or raising ω_m , both of which we examine in the sensitivity analysis.

4. Noise-Budget Summary		
Mechanism	Contribution	Status
Gas damping	$Q_{\text{gas}} \geq 10^9$ (10^{-9} mbar)	Negligible
Support / clamping loss	$Q_m = 10^8$ (vs $\sim 5 \times 10^8$ soft-clamped)	Limiting intrinsic loss
TLS losses	$Q \sim 10^8\text{--}10^9$ (strain-dependent)	Plausible source of gap
Thermoelastic damping	$Q_{\text{TED}} > 10^{10}$	Negligible
Photothermal	$P_{\text{abs}} \sim 10^{-17}$ W	Negligible
Zero-point motion	$x_{\text{zpf}} \approx 5.2 \times 10^{-15}$ m	Sets coupling scale
Thermal occupation	$n_{\text{th}} \approx 312$	Cooling required
Sideband regime	$\kappa/\omega_m = 1.5$ (outside RSB)	Cooling fundamentally degraded
Cooperativity	$C_0 \approx 2.7$	Below $C_0 > 10$ threshold
Quantum cooperativity	$C_q \approx 0.0086$	Thermal noise dominates

5. Comparison to Existing Experiments

Our baseline parameters are consistent with state-of-the-art SiN membrane-in-the-middle experiments at cryogenic temperatures. Microwave cQED platforms achieve much higher relative coupling but operate at GHz frequencies where ground-state preparation is straightforward.

Reference	Platform	Key parameters
O'Connell et al. 2010 [5]	Mech. resonator + qubit (μW)	$g_0/2\pi \sim 172$ Hz; $\omega_m/2\pi = 6$ GHz
Galland et al. 2014 [2]	SiN membrane (cryogenic)	$g_0/2\pi \sim 100$ Hz; resolved-sideband cooling
Riedinger et al. 2016 [3]	Optomechanical crystal	Entanglement via coherent drives
This note (baseline)	SiN membrane-in-middle (15 mK)	$C_0 \approx 2.7$; $\kappa/\omega_m = 1.5$; $n_{\text{th}} = 312$

6. Three Operational Regimes

It is essential to distinguish three operational regimes, since fidelity claims can be ambiguous if the regime is not specified:

Regime	Intracavity photons	Effective C	Swap fidelity	Notes
True single-photon	$n_c \approx 1$	$C = C_0 \approx 2.7$	$< 1\%$	Fundamentally limited by C_0
Linearized (drive)	$n_c \gg 1$	$C = n_c \cdot C_0 \gg 1$	Up to $\sim 100\%$	Quantum backaction dominates
Heralded (this paper)	$n_c \approx 1$ (signal)	Mixed	$\sim 20\%$ (optimistic)	Requires fast switching

The linearized-regime efficiency $\eta \approx 4C/(1+C)^2$ can approach unity at large C, but this represents amplification of a coherent state, not the swap of a single quantum [6]. True single-photon fidelity in the heralded regime is set by the bare cooperativity C_0 , not the linearized C, and remains below unity for our baseline.

7. Sensitivity Analysis

Parameter change	C_0 effect	C_q effect	Effort
$g_0 \rightarrow 2\times$ (mode-volume reduction)	$\times 4$	$\times 4$	Hard
$Q_m \rightarrow 2\times$ (better clamping)	$\times 2$	$\times 2$	Moderate
$F \rightarrow 2\times$ (better mirrors)	$\times 2$	$\times 2$	Moderate
$\omega_m \rightarrow 10\times$ (stiffer mode)	unchanged	$\times 10$	Moderate
$T \rightarrow T/10$ (sub-50 μK)	unchanged	$\times 10$	Very hard / non-standard

The sensitivity analysis identifies ω_m as the highest-leverage parameter: a $10\times$ increase in mechanical frequency improves C_q by $10\times$ while simultaneously moving the system into the resolved-sideband regime (since κ/ω_m drops from 1.5 to 0.15). Combined with moderate increases in g_0 , F , and Q_m , the full path to high-fidelity single-photon storage becomes accessible — though all four targets must be met in a single integrated device.

8. Experimental Roadmap

Phase 0: Device optimization (room temperature).

Characterize Q_m vs. clamping geometry. Design phononic-crystal supports. Verify optical mode-waist matching to membrane area (50 μm scale).

Phase 1: Marginally unresolved (baseline).

Install in dilution fridge. With $\kappa/\omega_m = 1.5$, sideband cooling is fundamentally limited; expected $n_{\min} \gtrsim 1$, not the resolved-sideband ideal of ~ 0.14 . Cavity ringdown adds ~ 100 ns switching overhead during which the mechanical state partially rethermalizes.

Phase 2: Resolved-sideband cavity.

Reduce $\kappa/2\pi < 500$ kHz (e.g. $F = 3 \times 10^4$ at $L = 1$ cm), bringing $\kappa/\omega_m < 0.5$. Demonstrate sideband cooling to $n_{\text{eff}} < 0.3$. This is the gating step.

Phase 3: Linearized-regime single-photon attempt.

Heralded SPDC source ($\sim 10^5 \text{ s}^{-1}$). Optimistic efficiency $\sim 20\%$ in the heralded regime; true single-photon fidelity remains $< 1\%$ until C_0 exceeds the threshold. Quantum process tomography on retrieved photons.

Phase 4: Aspirational target.

Raise ω_m to 10 MHz ($n_{\text{th}} \approx 31$). Set $\kappa/2\pi = 500$ kHz, $g_0/2\pi = 2$ kHz (projected), $Q_m = 10^9$ (projected). Predicted $C_q \approx 100$, enabling high-fidelity single-photon storage. Each individual target lies within the demonstrated capability of state-of-the-art labs; achieving all four simultaneously in a single device represents a significant integration challenge.

9. Conclusions

With the self-consistent baseline parameter set, single-photon storage is not achievable. The three structural obstacles are: (i) single-photon cooperativity $C_0 \approx 2.7$, below the $C_0 > 10$ threshold; (ii) thermal occupation $n_{\text{th}} \approx 312$, requiring active cooling; and (iii) operation outside the resolved-sideband regime ($\kappa/\omega_m = 1.5$), which fundamentally limits cooling efficiency. All three are quantitatively explained by standard optomechanical theory [1,4]. No unidentified loss channel or new physics is required.

The path forward is also clear. Raising ω_m by an order of magnitude is the highest-leverage move, simultaneously suppressing thermal occupation and entering the resolved-sideband regime. Combined with moderate improvements in g_0 , F , and Q_m , the same analysis predicts $C_q \approx 100$ and enables high-fidelity single-photon storage. **The baseline does not work for the reasons identified; the aspirational target is achievable in principle but represents a significant integration challenge.**

This note provides a platform-specific bound for SiN membrane-in-the-middle systems at telecom wavelengths. It does not address other architectures (microwave cQED, levitated particles, optomechanical crystals, hybrid magnon systems) where strong coupling is more readily achieved. The methodological value lies in the structured accounting: every limitation maps to a specific equation, every projected improvement maps to a specific parameter, and every claim is bounded by a stated regime of validity.

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Appendix: Numerical Estimates

Quantity	Formula	Value
Zero-point motion	$\sqrt{\hbar/(2m\omega_m)}$	$5.2 \times 10^{-15} \text{ m}$
Cavity linewidth	$\kappa/2\pi = c/(2LF)$	$1.5 \times 10^6 \text{ Hz}$
Mechanical damping	$\Gamma_m = \omega_m/Q_m$	0.063 rad/s
Cooperativity	$C_o = 4g_0^2/(\kappa\Gamma_m)$	2.67
Thermal occupation	$n_{th} \approx k_B T/(\hbar\omega_m)$	312
Quantum cooperativity	$C_q = C_o/n_{th}$	0.0086
Cavity storage time	$\tau_{cav} \approx 1/\kappa$	$1.06 \times 10^{-7} \text{ s}$
Single-photon Δx	$2g_0 x_{zpf} / \omega_m$	$1.0 \times 10^{-18} \text{ m}$
Sideband ratio	κ/ω_m	$1.50 \text{ (outside RSB)}$
Resolved-sideband floor (does not apply)	$(\kappa/4\omega_m)^2$	$0.14 \text{ (formal only)}$
Ground-state T (passive)	$\hbar\omega_m/k_B$	$\sim 48 \text{ }\mu\text{K}$

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